

REASONING SETTING

● OpenThinker3-7B vs parent
A · OPENTHINKER3-7B VS PARENT

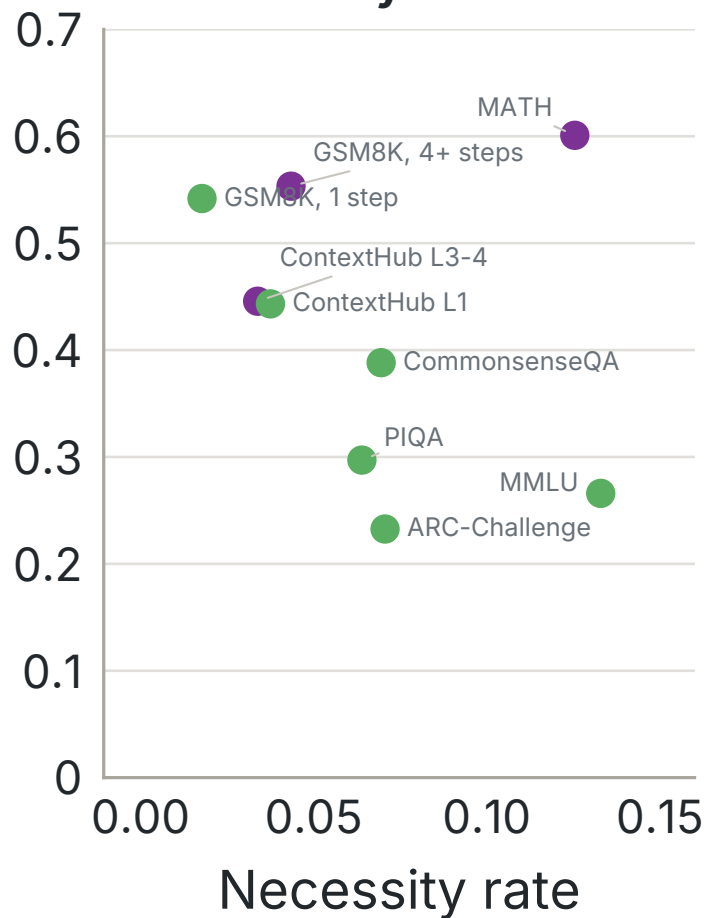
◆ Qwen3-8B, thinking on vs off
B · QWEN3-8B, THINKING ON VS OFF

CORPORA (A, B)

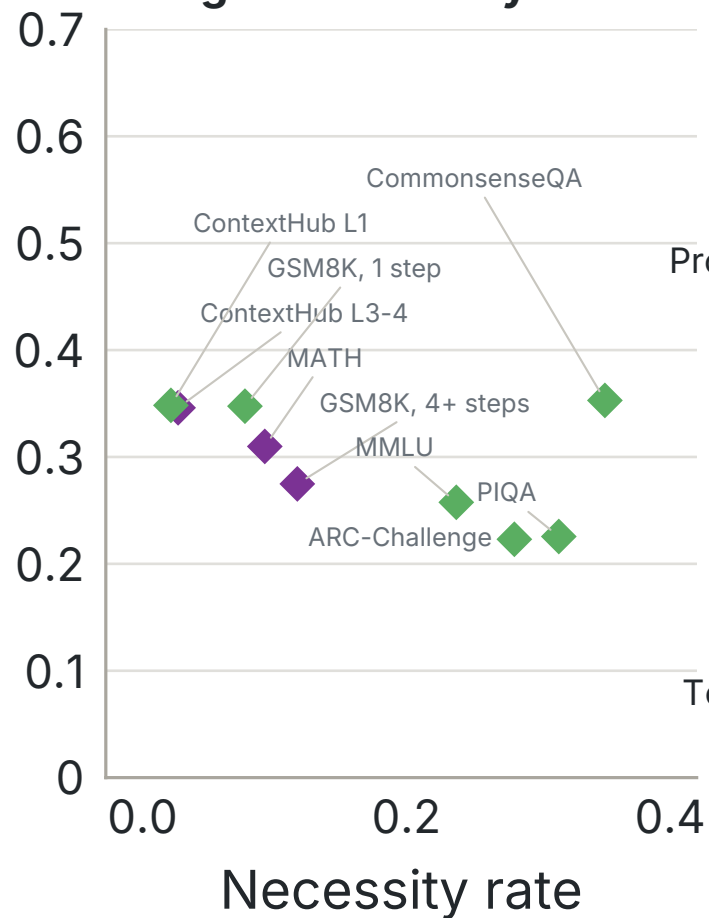
■ Needs-reasoning corpora ■ No-reasoning corpora
C · PER-QUESTION AUROC, 95% INTERVALS

Retrieval gain from end of thought ↑

Gain follows corpus type,
not necessity rate



Same on the thinking toggle,
at higher necessity rates



No map score identifies
which MATH questions need reasoning

